

Cycle-Moment Sinkhorn Flows: A Brockett-like View of Trace-Constrained Routing on the Birkhoff Polytope

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Abstract

We sketch a geometric flow on the positive interior of the Birkhoff polytope motivated by the traveling salesperson problem, Brockett’s double-bracket flow, and modern Sinkhorn and mirror-descent methods. The central object is a KL-gradient flow over doubly stochastic matrices augmented by trace-power constraints of the form

$$h_k(P) = \text{tr}(P^k) - b_k.$$

For TSP-like routing, these trace moments distinguish single Hamiltonian cycles from subtour decompositions. The resulting system is not proposed primarily as a competitive TSP solver, but as a mathematical object: a transport-polytope analogue of Brockett-style constrained matrix flows. We discuss its constrained geometry, primal-dual and augmented variants, local linearisation, Lyapunov structure, damping of trace-violating modes, degeneracy near the uniform doubly stochastic matrix, and reduced-cost stability near permutation vertices. Small numerical experiments illustrate the predicted stability phenomena.

1 Motivation

Brockett’s double-bracket flow gives a dynamical-systems view of sorting, diagonalisation, and constrained matrix evolution [3]. Classical Brockett-style flows usually live on isospectral or orthogonal/conjugacy manifolds, where commutator structure preserves constraints while a potential function drives the system toward an ordered state.

Here we ask for an analogous object on the Birkhoff polytope

$$\mathcal{B}_n = \{P \in \mathbb{R}_+^{n \times n} : P\mathbf{1} = \mathbf{1}, P^\top \mathbf{1} = \mathbf{1}\}.$$

The vertices of $\mathcal{B}_n$ are permutation matrices by the Birkhoff–von Neumann theorem. For directed TSP, a tour can be represented by a permutation matrix corresponding to a single n -cycle. The usual assignment relaxation allows all permutation matrices, including disjoint subtour decompositions. Trace powers provide a compact differentiable way to detect closed walks and subtours.

The motivating question is:

Can we define a Brockett-like flow on the Birkhoff polytope whose geometry is Sinkhorn/KL rather than orthogonal/commutator, and whose moment constraints favour Hamiltonian cycles?

The practical version is a differentiable Sinkhorn relaxation for TSP. The more interesting mathematical version is a constrained mirror-gradient flow with trace moments and a stability theory that may apply more generally to differentiable routing and assignment problems.

2 Birkhoff and trace-moment setup

Let $P \in \mathcal{B}_n^\circ$, where

$$\mathcal{B}_n^\circ = \{P > 0 : P\mathbf{1} = \mathbf{1}, P^\top \mathbf{1} = \mathbf{1}\}.$$

Let $C \in \mathbb{R}^{n \times n}$ be an edge-cost matrix. The linear assignment/TSP-like cost is

$$\langle C, P \rangle = \text{tr}(C^\top P).$$

Define trace-moment constraints

$$h_k(P) = \text{tr}(P^k) - b_k.$$

For a single directed Hamiltonian cycle, a natural target is

$$b_k = 0, \quad 1 \leq k < n, \quad b_n = n.$$

The constraints $\text{tr}(P^k) = 0$ for $k < n$ suppress shorter closed cycles, while $\text{tr}(P^n) = n$ is consistent with a single n -cycle permutation. Trace powers and related trace/exponential trace constraints are common in continuous acyclicity formulations [8]; recent work has also explored trace-guided continuous relaxations for ATSP [7].

Lemma 1 (Trace moments characterise Hamiltonian cycles). *Let $P \in \mathcal{B}_n$. Then P is the permutation matrix of a single directed n -cycle if and only if*

$$\text{tr}(P^k) = 0, \quad k = 1, \dots, n-1.$$

In that case, automatically $\text{tr}(P^n) = n$.

Proof. If P is a single n -cycle permutation matrix, then P^k has no fixed points for $1 \leq k < n$, so $\text{tr}(P^k) = 0$.

Conversely, suppose $P \in \mathcal{B}_n$ and $\text{tr}(P^k) = 0$ for $1 \leq k < n$. Since $P \geq 0$,

$$\text{tr}(P^k) = \sum_{i_1, \dots, i_k} P_{i_1 i_2} P_{i_2 i_3} \cdots P_{i_k i_1},$$

and every term is nonnegative. Hence $\text{tr}(P^k) = 0$ implies there is no positive-weight closed walk of length k .

By the Birkhoff–von Neumann theorem, write

$$P = \sum_{\ell} \alpha_{\ell} \Pi_{\ell}, \quad \alpha_{\ell} > 0, \quad \sum_{\ell} \alpha_{\ell} = 1,$$

where each Π_{ℓ} is a permutation matrix. If any Π_{ℓ} contained a cycle of length $m < n$, that cycle would give a positive term in $\text{tr}(P^m)$, a contradiction. Hence each Π_{ℓ} is a single n -cycle.

It remains to rule out a convex combination of distinct n -cycles. Relabel the vertices so that one such cycle is

$$1 \rightarrow 2 \rightarrow \cdots \rightarrow n \rightarrow 1.$$

If another n -cycle is distinct, it has some edge $i \rightarrow j$ with $j \neq i+1$ modulo n . Combining this edge with the path in the first cycle from j back to i gives a directed cycle of length strictly less than n in the support of P , again contradicting the trace constraints. Therefore all permutation matrices in the decomposition are the same n -cycle, and P is that permutation matrix. Finally $P^n = I$, so $\text{tr}(P^n) = n$. $\square$

Remark 1. *On an orthogonal or isospectral manifold, trace-power constraints mostly constrain eigenvalues and leave a large continuous orbit. On the nonnegative stochastic/Birkhoff side, trace powers have a direct closed-walk interpretation. This is what makes the trace constraints combinatorial.*

3 KL geometry on the Birkhoff polytope

A natural flow on $\mathcal{B}_n^\circ$ is the KL or entropic mirror-gradient flow. Given a scalar energy $E(P)$ with Euclidean gradient $G = \nabla_P E(P)$, the Birkhoff-preserving mirror flow can be written

$$\dot{P}_{ij} = -P_{ij}(G_{ij} - a_i - b_j),$$

where row and column potentials a, b are chosen so that $\dot{P}\mathbf{1} = 0$ and $\dot{P}^\top \mathbf{1} = 0$. This is related to Sinkhorn and mirror-flow viewpoints on transport polytopes [2, 4, 6].

Let

$$r_i = \sum_j P_{ij} G_{ij}, \quad c_j = \sum_i P_{ij} G_{ij}.$$

Preservation of row and column sums gives

$$\begin{aligned} a + Pb &= r, \\ P^\top a + b &= c, \end{aligned}$$

or

$$\begin{bmatrix} I & P \\ P^\top & I \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} r \\ c \end{bmatrix}.$$

There is a gauge freedom $a \mapsto a + \gamma \mathbf{1}$, $b \mapsto b - \gamma \mathbf{1}$.

The tangent space of the Birkhoff interior is

$$T_P \mathcal{B}_n^\circ = \{V \in \mathbb{R}^{n \times n} : V\mathbf{1} = 0, V^\top \mathbf{1} = 0\}.$$

The KL/Fisher metric on positive matrices is

$$\langle U, V \rangle_P = \sum_{ij} \frac{U_{ij} V_{ij}}{P_{ij}}.$$

Without row/column constraints, the KL-gradient of E is $P \odot G$, because

$$\langle P \odot G, V \rangle_P = \sum_{ij} G_{ij} V_{ij} = DE(P)[V].$$

The Birkhoff constraints subtract row and column potentials before multiplying by P .

Define

$$W_{ij} = P_{ij}(G_{ij} - a_i - b_j).$$

For any tangent direction $V \in T_P \mathcal{B}_n^\circ$,

$$\begin{aligned} \langle W, V \rangle_P &= \sum_{ij} (G_{ij} - a_i - b_j) V_{ij} \\ &= \sum_{ij} G_{ij} V_{ij} - \sum_i a_i \sum_j V_{ij} - \sum_j b_j \sum_i V_{ij} \\ &= DE(P)[V]. \end{aligned}$$

The potential terms vanish because tangent directions have zero row and column sums. Hence $W = \text{grad}_{\mathcal{B}}^{\text{KL}} E(P)$, and the negative KL-gradient flow is $\dot{P} = -W$.

Strict positivity follows from the multiplicative form. Each entry satisfies $\dot{P}_{ij} = P_{ij} R_{ij}(t)$, so

$$P_{ij}(t) = P_{ij}(0) \exp \left(\int_0^t R_{ij}(s) ds \right),$$

and positive entries remain positive while the vector field remains finite.

4 Cycle-moment Sinkhorn flow

Define the augmented trace-penalty energy

$$E_\rho(P) = \langle C, P \rangle + \frac{\rho}{2} \sum_{k=1}^n h_k(P)^2.$$

Then

$$\nabla_P E_\rho(P) = C + \rho \sum_{k=1}^n h_k(P) k(P^{k-1})^\top.$$

The penalty flow is therefore

$$\dot{P}_{ij} = -P_{ij}(G_{ij} - a_i - b_j).$$

This is the continuous-time analogue of exponentiated-gradient descent followed by Sinkhorn projection.

5 Trace derivatives and constraint Jacobian

Let $f_k(P) = \text{tr}(P^k)$. Expanding to first order,

$$(P + \epsilon \delta P)^k = P^k + \epsilon \sum_{r=0}^{k-1} P^r \delta P P^{k-1-r} + O(\epsilon^2).$$

Taking traces and using cyclicity gives

$$Df_k(P)[\delta P] = k \text{tr}(P^{k-1} \delta P).$$

Thus

$$\nabla_P \text{tr}(P^k) = k(P^{k-1})^\top.$$

The trace-constraint Jacobian $A = Dh(P)$ has rows

$$A_k[\delta P] = k \text{tr}(P^{k-1} \delta P).$$

Differentiating again gives the directional second variation

$$D^2 f_k(P)[\delta P, \delta P] = k \sum_{r=0}^{k-2} \text{tr}(P^r \delta P P^{k-2-r} \delta P).$$

The linear cost contributes no Hessian; local curvature of the trace-constrained Lagrangian is generated by the trace terms.

6 Primal-dual and augmented variants

Introduce live multipliers λ_k and define

$$\mathcal{L}(P, \lambda) = \langle C, P \rangle + \sum_{k=1}^n \lambda_k h_k(P).$$

The primal-dual mirror flow is

$$\dot{P} = -\text{grad}_{\mathcal{B}}^{\text{KL}} \mathcal{L}(P, \lambda), \quad \dot{\lambda}_k = \eta h_k(P).$$

The augmented version is

$$\mathcal{L}_\rho(P, \lambda) = \langle C, P \rangle + \lambda^\top h(P) + \frac{\rho}{2} \|h(P)\|^2.$$

The augmented term is not merely a penalty; locally it damps trace-visible primal-dual modes, as described below.

7 Lyapunov structure

For any smooth energy E and KL-gradient flow $\dot{P} = -W$, where $W = \text{grad}_{\mathcal{B}}^{\text{KL}} E(P)$,

$$\frac{dE}{dt} = DE(P)[\dot{P}] = \langle W, -W \rangle_P = -\|W\|_P^2 \leq 0.$$

In coordinates,

$$\|W\|_P^2 = \sum_{ij} P_{ij} (G_{ij} - a_i - b_j)^2.$$

Thus for the pure penalty flow,

$$\frac{dE_\rho}{dt} = -\sum_{ij} P_{ij} (G_{ij} - a_i - b_j)^2 \leq 0.$$

The pure primal-dual flow does not satisfy the same identity for $\mathcal{L}$. Indeed,

$$\frac{d\mathcal{L}}{dt} = D_P \mathcal{L}[\dot{P}] + h(P)^\top \dot{\lambda},$$

and with $\dot{\lambda} = \eta h(P)$ the second term is $\eta \|h(P)\|^2 \geq 0$. This is consistent with the oscillatory modes of the primal-dual linearisation.

8 Local linearisation and damping

Let (P_*, λ_*) be an interior stationary point and let $A_* = Dh(P_*)$. Schematically, the primal-dual linearisation has block form

$$\begin{bmatrix} \delta \dot{P} \\ \delta \dot{\lambda} \end{bmatrix} = \begin{bmatrix} -M_* H_* & -M_* A_*^\top \\ \eta A_* & 0 \end{bmatrix} \begin{bmatrix} \delta P \\ \delta \lambda \end{bmatrix}.$$

Here M_* denotes the KL mobility/projection operator at P_* . Concretely, for a Euclidean covector Q , $M_* Q$ is the Birkhoff-tangent vector

$$(M_* Q)_{ij} = P_{*,ij} (Q_{ij} - \alpha_i - \beta_j),$$

where α, β are chosen so that $M_* Q$ has zero row and column sums.

For the augmented primal-dual flow, the schematic linearisation becomes

$$\begin{bmatrix} \delta \dot{P} \\ \delta \dot{\lambda} \end{bmatrix} = \begin{bmatrix} -M_* (H_* + \rho A_*^\top A_*) & -M_* A_*^\top \\ \eta A_* & 0 \end{bmatrix} \begin{bmatrix} \delta P \\ \delta \lambda \end{bmatrix},$$

with the caveat that the intrinsic metric-aware expression involves M_* , for example through $A_* M_* A_*^\top$ in the dual trace space.

For the clean trace-mode picture, use local coordinates z on the Birkhoff tangent space and linearise $h(P_* + z) \approx Az$. Ignoring cost curvature, the pure primal-dual local model is

$$\dot{z} = -A^\top \lambda, \quad \dot{\lambda} = \eta Az.$$

Let $Av_i = \sigma_i u_i$ be a singular mode and write $z = qv_i$, $\lambda = pu_i$. Then

$$\dot{q} = -\sigma_i p, \quad \dot{p} = \eta \sigma_i q,$$

so

$$\ddot{q} + \eta \sigma_i^2 q = 0.$$

This is the linearised ringing mechanism.

For the augmented system,

$$\dot{z} = -A^\top \lambda - \rho A^\top Az, \quad \dot{\lambda} = \eta Az.$$

The same singular mode satisfies

$$\ddot{q} + \rho \sigma_i^2 \dot{q} + \eta \sigma_i^2 q = 0.$$

Thus augmentation adds damping coefficient $\rho \sigma_i^2$ in each trace-visible singular mode. In the full nonlinear Sinkhorn parameterisation this undamped control structure may appear not as a clean sinusoid, but as delayed high-residual plateaus followed by sharp support changes.

9 Uniform initialisation degeneracy

Let

$$P_0 = \frac{1}{n} \mathbf{1} \mathbf{1}^\top.$$

Then $P_0^k = P_0$ for $k \geq 1$. For $\delta P \in T_{P_0} \mathcal{B}_n$, and $k \geq 2$,

$$D \operatorname{tr}(P^k)|_{P=P_0}[\delta P] = k \operatorname{tr}(P_0 \delta P) = \frac{k}{n} \mathbf{1}^\top \delta P \mathbf{1} = 0.$$

Thus all trace-power constraints with $k \geq 2$ are first-order invisible at the uniform Birkhoff point. The case $k = 1$ is special:

$$D \operatorname{tr}(P)[\delta P] = \operatorname{tr}(\delta P),$$

which is not automatically zero on the Birkhoff tangent space. If diagonal/self-loop entries are hard-masked, this remaining first-order signal is removed. Consequently, in the no-self-loop setting, exact uniform initialisation is degenerate for trace penalties; small random logit perturbations are needed to reveal trace-gradient information.

10 Boundary behaviour and reduced costs

Permutation tours lie on the boundary of $\mathcal{B}_n$. The KL flow degenerates at zeros: if $P_{ij} = 0$, then $\dot{P}_{ij} = 0$. Boundary behaviour is therefore best interpreted through near-boundary limits or log coordinates.

For any positive entry,

$$\frac{d}{dt} \log P_{ij} = -(G_{ij} - a_i - b_j).$$

Define the reduced augmented cost

$$\Delta_{ij} = G_{ij} - a_i - b_j.$$

Then

$$\frac{d}{dt} \log P_{ij} = -\Delta_{ij}.$$

Near a permutation tour, off-support entries are small but positive if we remain in the Birkhoff interior. Positive reduced cost suppresses such an edge; negative reduced cost makes it grow.

A single off-support edge is not generally a feasible perturbation. Feasible perturbations of a permutation vertex preserve row and column sums and are generated by alternating-cycle exchange directions in the assignment polytope [1]. If D is such a feasible exchange direction, then $D\mathbf{1} = 0$ and $D^\top \mathbf{1} = 0$. Consequently,

$$\langle \Delta, D \rangle = \sum_{ij} (G_{ij} - a_i - b_j) D_{ij} = \langle G, D \rangle,$$

because the row/column potentials vanish on feasible directions. Hence a candidate tour is locally stable against a feasible exchange direction D when

$$\langle G, D \rangle > 0.$$

This is a reduced-cost diagnostic in the sense of linear programming and assignment optimisation [5], not a global optimality theorem.

11 Practical autodiff/Sinkhorn recipe

The practical differentiable relaxation uses logits X and a temperature τ :

$$P = \text{Sinkhorn}(\exp(X/\tau)).$$

Then optimise

$$E_{\rho, \tau}(X) = \langle C, P \rangle + \frac{\rho}{2} \sum_{k=1}^n h_k(P)^2.$$

The stability analysis suggests:

1. do not initialise exactly at $X = 0$ in the no-self-loop setting;
2. use small random logits to break symmetry;
3. increase ρ gradually so trace-visible modes are damped without making the system excessively stiff;
4. decrease τ gradually to encourage collapse toward permutation vertices;
5. monitor reduced costs $\Delta_{ij} = G_{ij} - a_i - b_j$ near convergence.

12 Numerical investigation

The experiments are diagnostic rather than benchmark-oriented. They test whether the stability predictions appear in small controlled instances.

12.1 Circle sanity check

On points arranged on a circle, the continuation schedule recovers the expected geometric tour. A fixed penalty can produce a valid cycle with poor geometry, demonstrating the separation between validity and optimality.

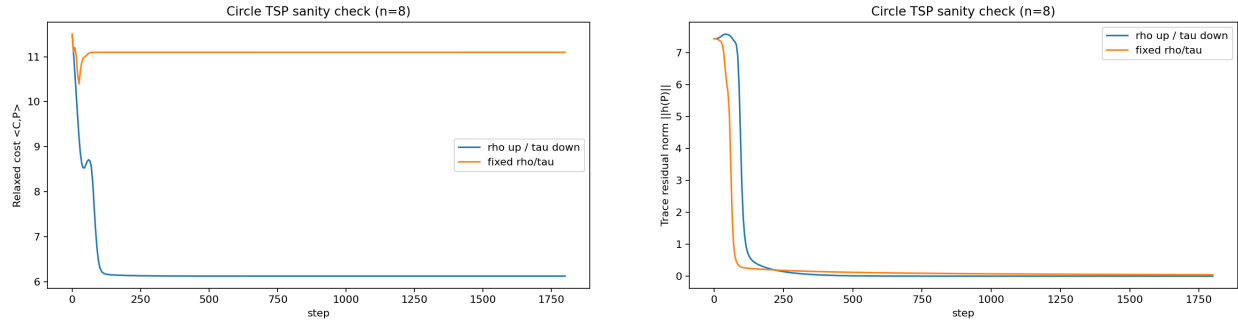


Figure 1: Circle sanity check. The scheduled run reaches the optimal geometric tour cost while managing trace residuals.

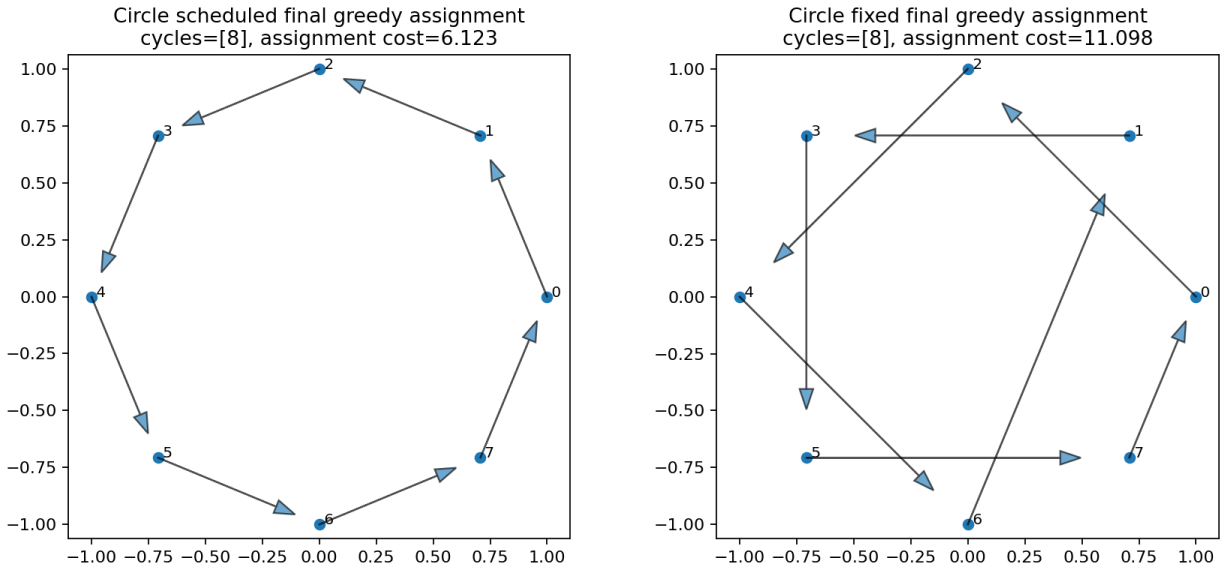


Figure 2: Circle final assignments. The scheduled run recovers the nearest-neighbour tour; the fixed run can satisfy Hamiltonicity while selecting a poor crossing/chord tour.

12.2 Two-cluster subtour pressure

The two-cluster instance is designed so the cost favours two within-cluster subtours. Low trace pressure gives cheap cycle covers such as $[4, 4]$; higher trace pressure forces a single 8-cycle.

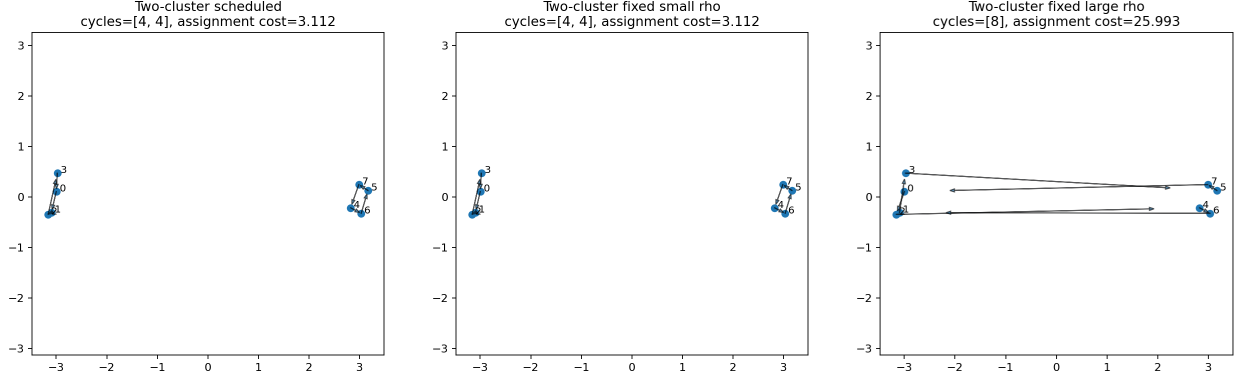


Figure 3: Two-cluster assignments. Scheduled and small- ρ runs remain in subtour covers; large ρ forces a single cycle.

12.3 Rho-pressure transition

A fixed- ρ sweep makes the Hamiltonicity-pressure transition explicit.

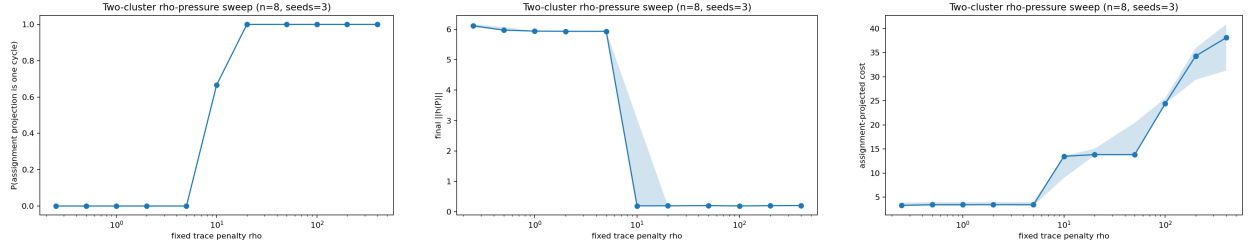


Figure 4: Rho-pressure sweep. Increasing ρ collapses trace residuals and turns subtour covers into single-cycle assignments, but assignment-projected cost rises once Hamiltonicity is forced.

12.4 Rho/tau phase diagram

The phase diagram shows that ρ primarily controls the subtour-to-Hamiltonian transition, while τ controls entropy and sharpness.

12.5 Uniform degeneracy and ringing

The uniform diagnostic confirms first-order trace-gradient collapse at exact uniform initialisation in the no-self-loop setting. The linearised top-mode experiment shows pure primal-dual ringing and augmented damping.

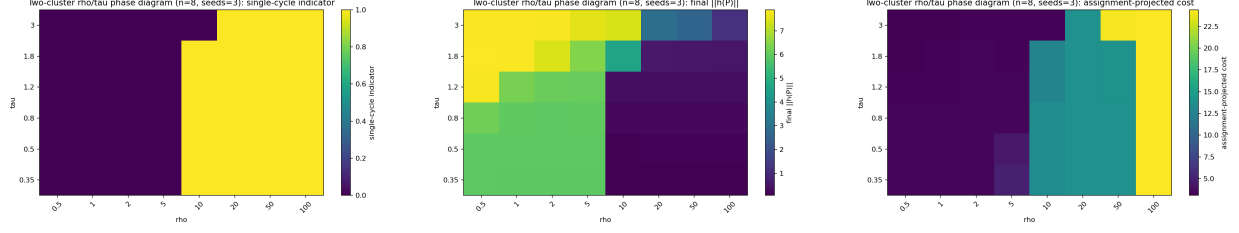


Figure 5: Two-cluster phase diagram. Trace pressure controls the transition to single-cycle assignments; assignment cost can rise sharply in the Hamiltonian phase.

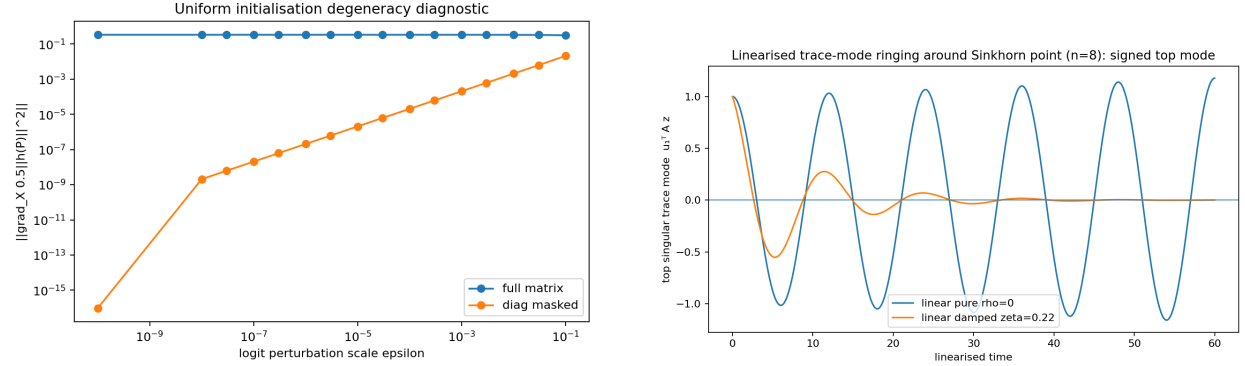


Figure 6: Left: uniform initialisation degeneracy. Right: linearised top trace mode, showing undamped primal-dual oscillation and augmented damping.

12.6 Nonlinear primal-dual snapping

In the full nonlinear Sinkhorn/logit system, pure primal-dual dynamics often appears as delayed high-residual plateaus followed by snapping rather than a clean sinusoid. Augmentation damps the trace-visible modes directly.

13 Relation to Brockett flow

The analogy with Brockett’s double-bracket flow is structural rather than literal. Classical Brockett-style flows use orthogonal/isospectral geometry, commutator projection, and sorting/diagonalisation by a potential. The present flow uses Birkhoff/transport-polytope geometry, Sinkhorn/KL projection via row-column potentials, and cycle-moment shaping by a routing cost. In Brockett flow, commutators remove forbidden normal directions on an isospectral orbit. Here, row/column potentials remove forbidden marginal directions on the Birkhoff polytope.

A concise description is: a transport-polytope analogue of Brockett flow, where isospectral commutator geometry is replaced by KL/Sinkhorn geometry, and Hamiltonian cycles arise as trace-moment constrained boundary strata.

14 Discussion and open questions

The experiments support the stability analysis rather than a solver-performance claim. The main picture is:

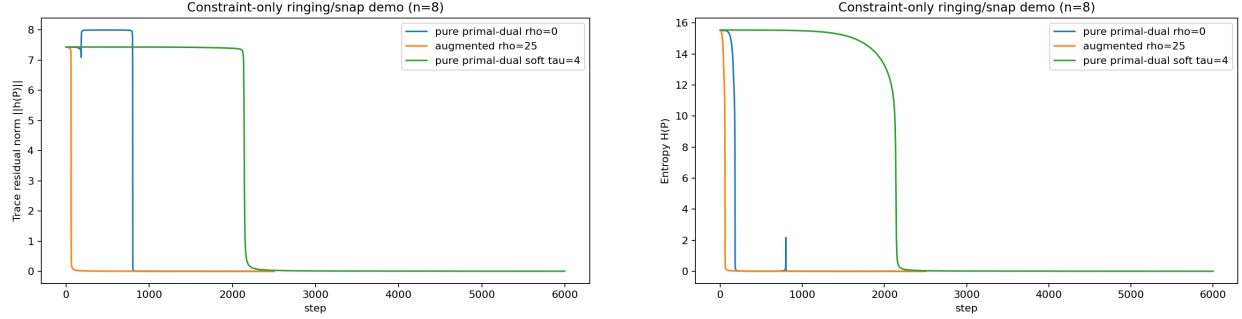


Figure 7: Nonlinear primal-dual diagnostics. The full dynamics can show delayed collapse and support hardening rather than smooth small-signal ringing.

1. continuation can recover an obvious geometric tour;
2. fixed trace pressure can produce valid but poor tours;
3. clustered instances reveal cheap subtour attractors;
4. increasing ρ produces a transition from subtours to single cycles;
5. exact uniform initialisation is first-order degenerate under no-self-loop trace constraints;
6. linearised primal-dual trace modes ring without augmentation;
7. augmented penalties damp trace-visible modes.

Open questions include: the best citation lineage for the trace/Hamiltonian characterisation; the exact rank of the trace-constraint Jacobian at symmetric points; adaptive tuning of ρ using the spectrum of AMA^T ; a rigorous reduced-cost stability theorem for alternating-cycle exchanges; and extensions to rectangular transport polytopes, partial matchings, capacity-constrained routing, and sparse attention maps.

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